

Indian Institute of Technology Bombay

Embedded Systems Design (EE712), 2026

Course Project Proposal

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1. Project Title

Real-Time Driver Assistance System using Intelligent Decision Making based on Depth Calculation and Image Segmentation

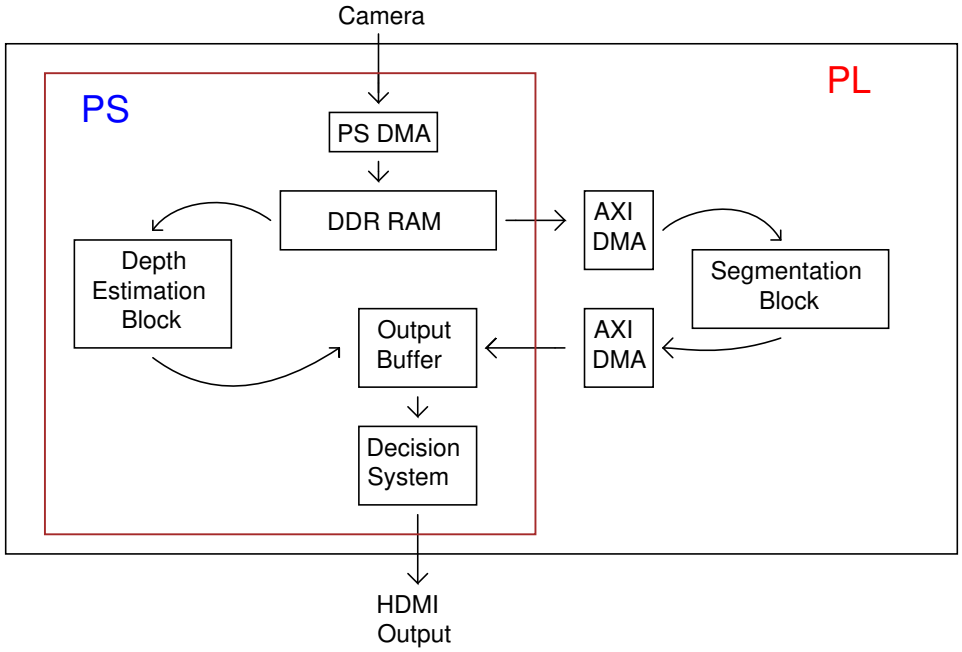
2. Team Members

S.No.	Name	Roll Number
1	Amol Pagare	22B3971
2	Arnav Agarwal	22B3917
3	Neil George	22B3963
4	Mitul Tandon	22b0305
5	Tanish Raghute	22B3974

3. Objectives and Goal of the Course Project

We wish to implement a real-time Advanced Driver Assistance System (ADAS) using the PYNQ-Z2 development board by integrating stereoscopic OR ML-based depth estimation with semantic segmentation. Utilizing a binocular vision setup via dual USB webcams (or a single camera if we face issues), the system will compute depth maps and identify critical environmental features such as lanes and obstacles via segmentation. To ensure real-time responsiveness, segmentation will be offloaded to the DPU unit or a custom IP on the PL, while the PS runs depth estimation. Using buffered outputs from these processes, a decision making algorithm (hardcoded or lightweight model) will trigger driving commands, such as emergency braking or speed adjustments.

4. Schematic/Block Diagram and Description



The proposed architecture implements an ADAS perception pipeline using the Processing System (PS) and Programmable Logic (PL). Image frames from the camera are transferred to DDR memory through the PS DMA, which acts as the central frame buffer. From DDR RAM, the data is processed through two parallel pipelines. In the PS, a depth estimation block computes distance information from the frames and stores the results in an output buffer. Simultaneously, frames are transferred to the PL through an AXI DMA, where the segmentation block performs object detection and classification. The segmentation results are then returned to the PS through another AXI DMA and stored in the output buffer. A decision system running on the PS combines the depth and segmentation information to determine the appropriate vehicle action. Finally, the processed output along with the decision is displayed through the HDMI interface.

5. MCU/Evaluation Board to be Used

PYNQ-Z2 Development Board

6. List of Components to be Used (BOM)

Components Available in WEL:

- HDMI + Monitor

Components to be Procured:

Component Name	**Quantity**	**Estimated Cost (INR)**
USB Webcams (Identical models for stereo vision)	2	1500
USB Hub (if required for dual-camera interfacing)	1	500
Total		**3000**

7. Timeline/Planning of the Course Project

Week	Task	Assigned To
1	Camera interfacing using DMA and baseline FPS check	All
	Finding and Comparing compatible Segmentation Models	All
2-3	Dual camera interfacing	Amol, Mitul, Arnav
	Running segmentation on video	Neil, Arnav, Tanish
	Depth algorithm development	Mitul, Neil, Tanish
	Decision-making algorithm development (3)	All
4-5	Integration of verticals & Debugging	All
6	Optimizing segmentation model choices (e.g., quantizing)	Neil, Arnav
	Performance benchmarking (FPS, Latency, and Power)	Tanish, Amol
	Resource utilization analysis	Mitul, Tanish
	Final project report and demo video recording	All

Table 1: Project Timeline and Work Distribution

References

[1] Z. Zhu, "Stereo Vision FPGA System," GitHub repository. Available: https://github.com/zhutmost/stereo-vision-fpga

[2] "Real-Time Stereo Vision System for Depth Estimation on FPGA," IEEE Xplore. Available: https://ieeexplore.ieee.org/document/10537163/

[3] L. Yang et al., "Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data," Available: https://depth-anything.github.io/

[4] A. Araújo, "DPU Implementation," GitHub repository. Available: https://github.com/andre1araujo/YOLO-on-PYNQ-Z2/tree/main/DPU%20implementation